# ECA5403 – CO1

## 1. Transformation between Cartesian, Cylindrical and spherical coordinate system

**Aim:**

To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using MATLAB.

**Program:**

**Test case (i)**

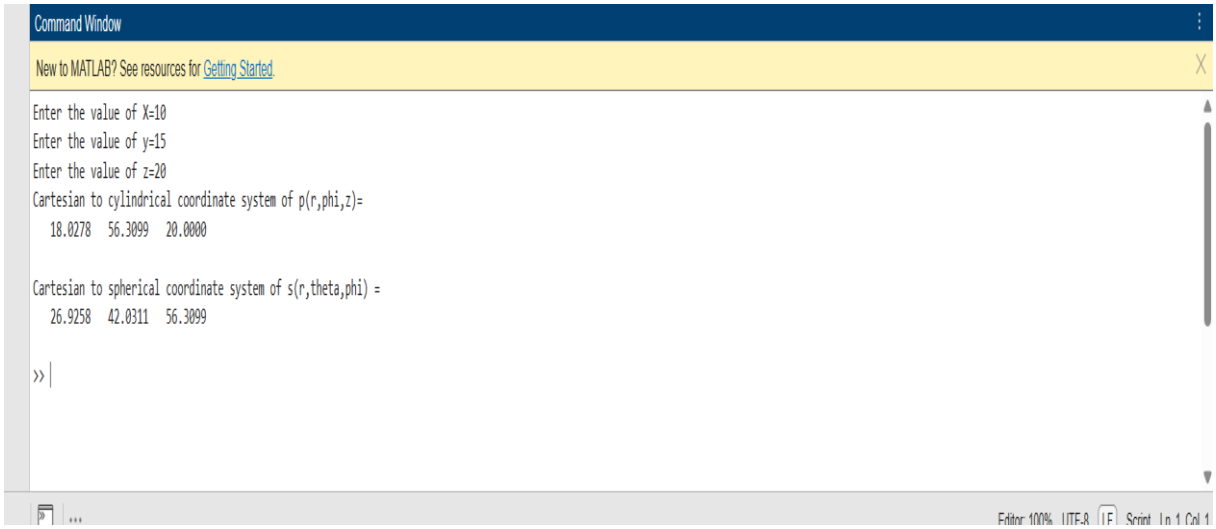
Transformation of Cartesian coordinate points in to Cylindrical and spherical coordinate system

```
clc;
x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');

% Cartesian to Cylindrical
r=sqrt(x^2+y^2);
phi=atan2d(y,x);
disp('Cartesian to cylindrical coordinate system of p(r,phi,z)=');
disp([r phi z]);

% Cartesian to Spherical
R=sqrt(x^2+y^2+z^2);
theta=acosd(z/R);
phi=atan2d(y,x);
disp ('Cartesian to spherical coordinate system of s(r,theta,phi) =');
disp ([R theta phi]);
```

**OUTPUT**:

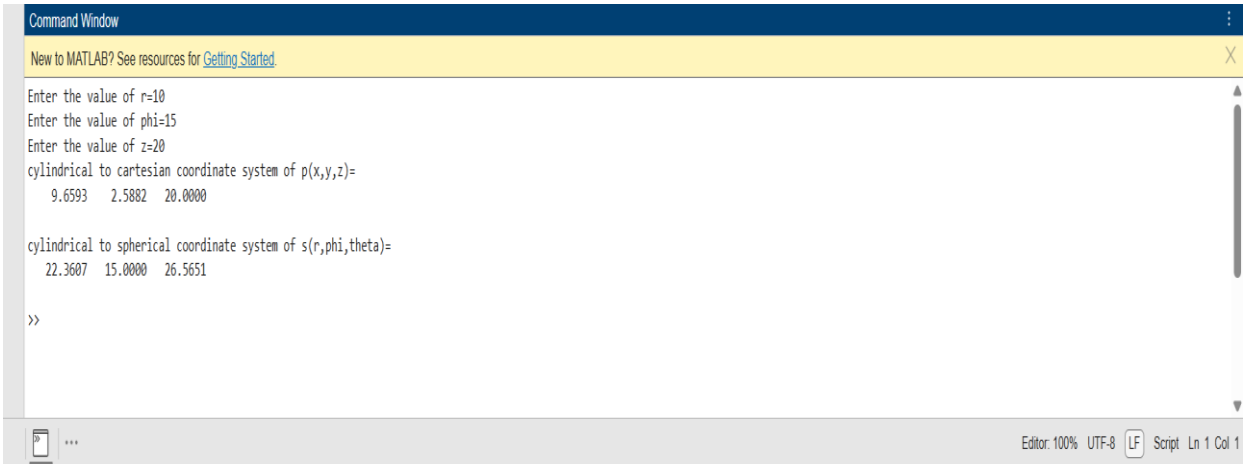



**Test case (ii)** Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

**Program:**

```
clc;
r=input('Enter the value of r=');
phi=input('Enter the value of phi=');
z=input('Enter the value of z=');
% Cylindrical to Cartesian
x=r*cosd(phi);
y=r*sind(phi);
disp('cylindrical to cartesian coordinate system of p(x,y,z)=');
disp([x y z]);
% Cylindrical to Spherical
R=sqrt(r^2+z^2);
theta=acosd(z/R);
disp('cylindrical to spherical coordinate system of s(r,phi,theta)=');
disp([R phi theta]);
```

**OUTPUT:**




**Test case (iii)**Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

**Program:**

```
clc;
r=input('Enter the value of r=');
phi=input('Enter the value of phi=');
theta=input('Enter the value of theta=');
% Spherical to Cartesian
x=r*sind(theta)*cosd(phi);
y=r*sind(theta)*sind(phi);
z=r*cosd(theta);
disp('spherical to cartesian coordinate
system of p(x,y,z)=');
disp([x y z]);
% Spherical to Cylindrical
rho=r*sind(theta);
disp('spherical to cylindrical
coordinate system of s(rho,phi,z)=');
disp([rho phi z]);
```

**OUTPUT:**




**Result:**

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its classified two categories (a) Cartesian to cylindrical and spherical coordinate system (b) Cylindrical to cartesian and spherical coordinate system (c) spherical to cartesian and cylindrical coordinate system using MATLAB was successfully

| S. No. | Objective | Assessment Criterion | Excellent – 4 | Good – 3 | Satisfactory – 2 | Needs Improvement – 1 | Weighting (%) | Marks given | Marks obtained |
|---|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Apply coordinate-system definitions and angle conventions | Apply coordinate-system definitions, symbols, units, and angle conventions | Applies coordinate-system definitions and angle conventions throughout. | Uses correct definitions with one minor notation or convention error. | Shows partial understanding; multiple corrections are needed. | Uses incorrect coordinate definitions or angle conventions. | 25 | 2.5 | |
| 2 | Implement transformations in MATLAB | Implement six transformations in MATLAB | Implements all six transformations correctly using suitable degree-based functions. | Implements most transformations correctly with minor coding errors. | Implements some transformations but needs assistance to correct errors. | Unable to implement the required transformations. | 25 | 2.5 | |
| 3 | Verify numerical values and reverse transformations | Verify numerical values and reverse transformations | All outputs agree with analytical values; reverse transformations recover the original point within rounding precision. | Most values agree; small unexplained differences occur. | Several values differ or reverse verification is incomplete. | Outputs are incorrect or not verified. | 30 | 3 | |
| 4 | Interpret and document results | Interpret and document results | Presents complete tables, units, deviations, and a clear conclusion. | Documentation is clear with minor omissions. | Tables or interpretation are incomplete. | Results and conclusion are missing or unclear. | 20 | 2 | |
| Total | | | | | | | 100 | 10 | |